

BANK MANAGEMENT SYSTEM

PROJECT REPORT FILE

1. Introduction

The Bank Management System is a Python-based application developed to demonstrate basic banking operations through a simple menu-driven interface. It provides separate Admin and User modules for managing customer accounts, transactions, loans, feedback and passbook records. The project does not require MySQL; Python dictionaries and lists are used for temporary data storage.

2. Objectives

- Provide a simple banking interface.
- Implement separate Admin and User authentication.
- Manage customer account information.
- Perform deposit and withdrawal operations.
- Maintain transaction and passbook records.
- Manage loans and identify defaulters.
- Visualize loan trends using graphs.
- Demonstrate practical Python programming concepts.

3. Technologies Used

Language: Python 3

Libraries: Matplotlib, datetime and random

Data Storage: Python dictionaries and lists

Interface: Menu-driven console interface

4. System Modules

Admin Login: Authenticates the administrator.

User Login: Authenticates customers.

Account Management: View and update customer details.

Transaction Management: Deposit, withdraw and track transactions.

Loan Management: Add, update and monitor loans.

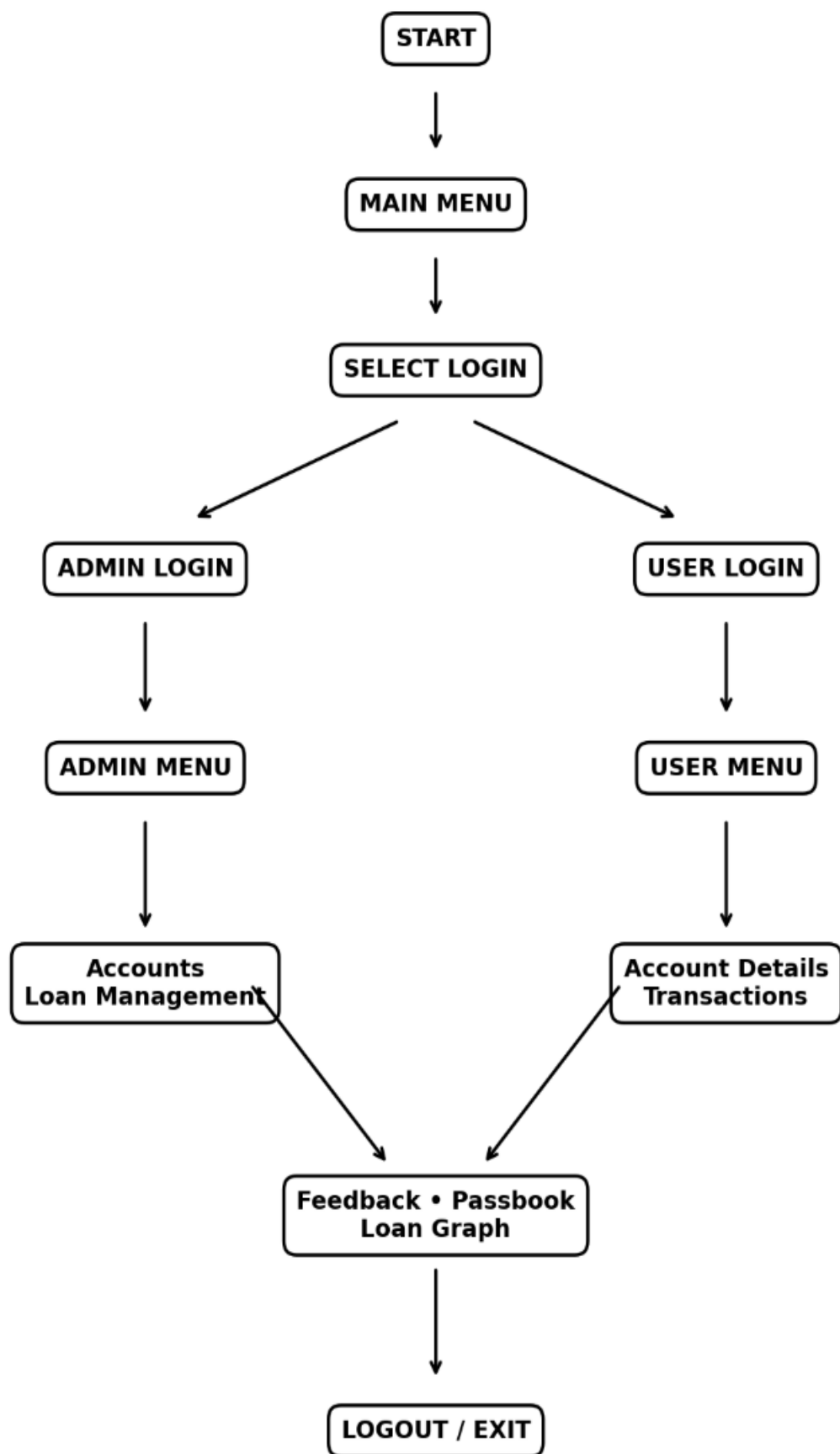
Feedback: Submit and view customer feedback.

Passbook: Display account transactions.

Data Visualization: Display yearly loan trends.

5. System Flowchart

Figure 1: Overall flow of the Bank Management System



6. Data Structures Used

The system uses nested dictionaries for account and loan information. Lists of dictionaries are used for feedback and transaction records. This makes the project lightweight and removes the need for an external database.

7. Main Features

- Pre-stored customer accounts and credentials.
- Automatic account number generation.
- Automatic transaction ID generation.
- Input validation.
- Balance updates after transactions.
- Loan status and unpaid-month tracking.
- Graphical loan-data visualization.

8. Sample Login Credentials

Admin: admin / admin123

Rahul: rahul / 1234

Priya: priya / 5678

Aman: aman / 1111

9. Advantages

- Easy to understand and operate.
- No database installation required.
- Demonstrates practical Python concepts.
- Separate Admin and User operations.
- Includes transaction tracking and visualization.

10. Limitations

Data is stored only while the program is running and is lost when the program closes. The system is designed as an academic project and does not include production-level security, encryption or permanent database storage.

11. Future Scope

- Add SQLite or MySQL for permanent storage.
- Develop a GUI using Tkinter.
- Add password hashing and stronger authentication.
- Add account-to-account fund transfers.
- Generate downloadable statements.
- Add advanced analytics and reports.

12. Conclusion

The Bank Management System demonstrates how Python can be used to model basic banking operations. It combines functions, dictionaries, lists, loops, conditional statements, input validation, random number generation, date and time handling, and Matplotlib visualization to create a practical academic application.